



Study of adhesin-substrate interactions at the molecular level for the development of better anti-bacterial materials

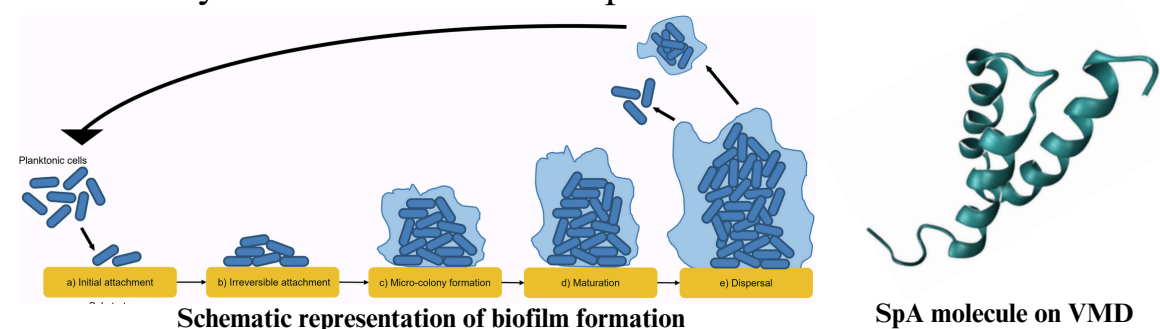


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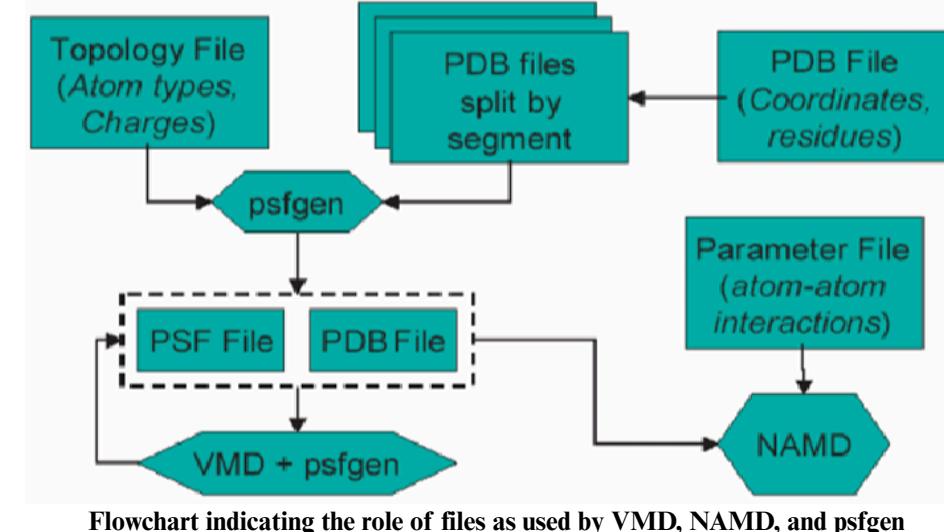
MOTIVATION

A study at MIMS, Karnataka reported a **4.43%** surgical site infection rate for **orthopedic implant surgery**, of which **60%** were caused by *S. aureus*. Both hydroxyapatite and mesoporous SiO₂ have been widely used as implant materials due to their high compatibility with human tissue and reduced risk of infection. The primary problem to be addressed was the process of anti-bacterial **biofilm formation** of SpA (key surface protein and virulence factor of *S. Aureus*) on the coating of an implant, initiated by **adhesion** of bacterial proteins onto the substrate.

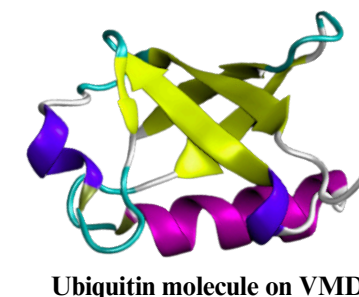


SYSTEM SETUP

Simulations required **NAMD 3.0.1** with **CUDA GPU** acceleration, and **VMD 1.9.3** was used for carrying out trajectory analysis and visualization. The **CHARMM27** force field was used for protein interactions and **Interface force field (IFF)** files were intended to be used for modelling the substrates. The **PSF** were generated using the **psfgen** in VMD from the **PDB**.



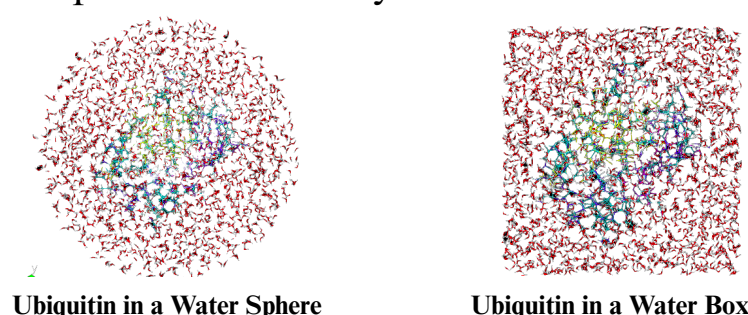
Due to the **biological complexity** of SpA, the host cell's protein regulatory mechanism first needed to be tested on a simpler system. Ubiquitin was chosen due to its structural simplicity.



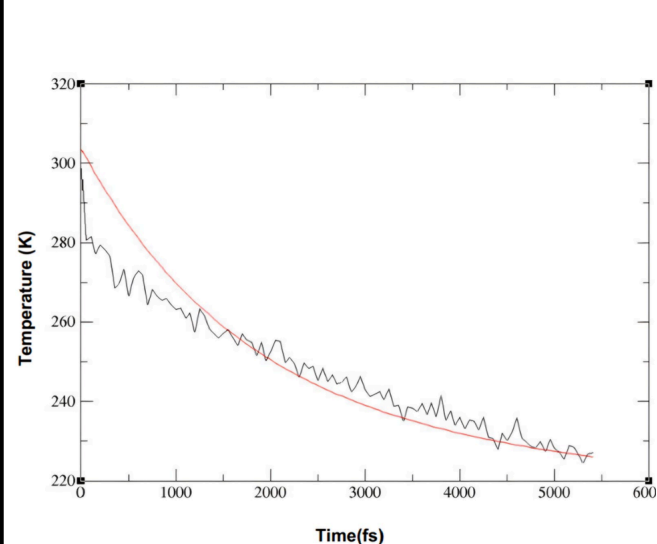
NAMD SIMULATION RESULTS ON UBIQUITIN

MINIMIZATION AND SOLVATION

Ubiquitin was solvated in two different environments i.e., a water box and a water sphere. Both systems underwent **100 steps** of energy minimization with velocities reset to **zero** prior to the equilibration step. Equilibration steps went for 2500 steps each and the system was heated to 310K.



COOLING BEHAVIOR

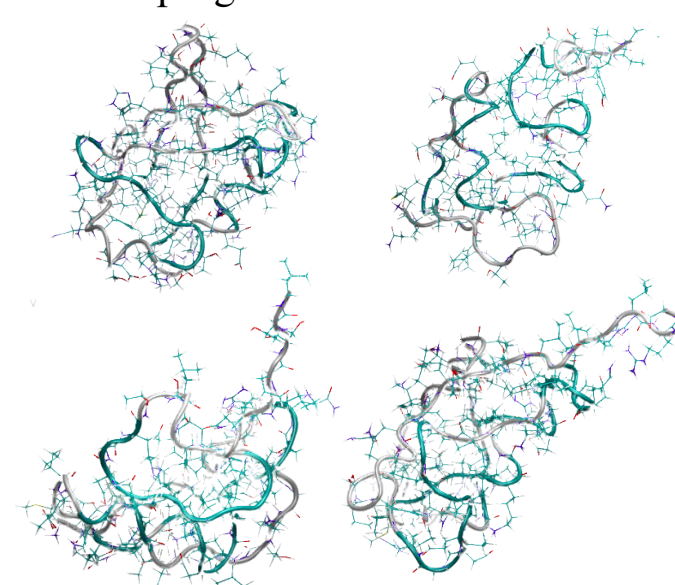


Exponential decay approximately from **310 K** to around **230 K** over about **5500 fs**, aligns with **Newton's law of cooling**.

It predicts if SpA re-equilibrates or denatures under thermal or mechanical stress.

STEERED MOLECULAR DYNAMICS (SMD)

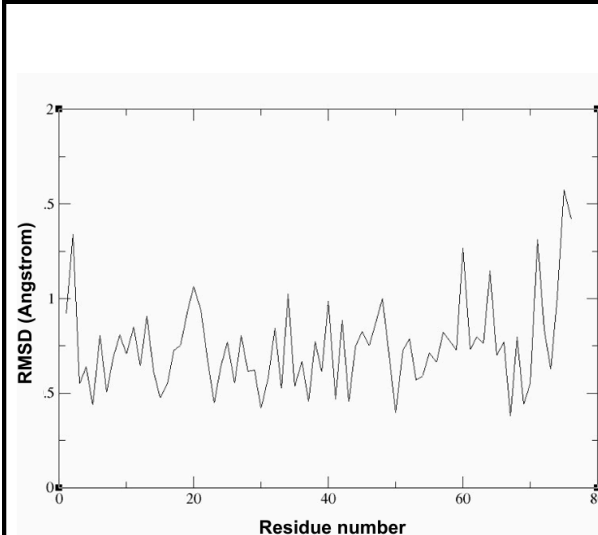
SMD simulations allow the investigation of molecular responses to forces by applying **controlled external forces to certain atoms**, referred to as SMD atoms, while keeping other atoms fixed.



Gradual unfolding of ubiquitin in SMD

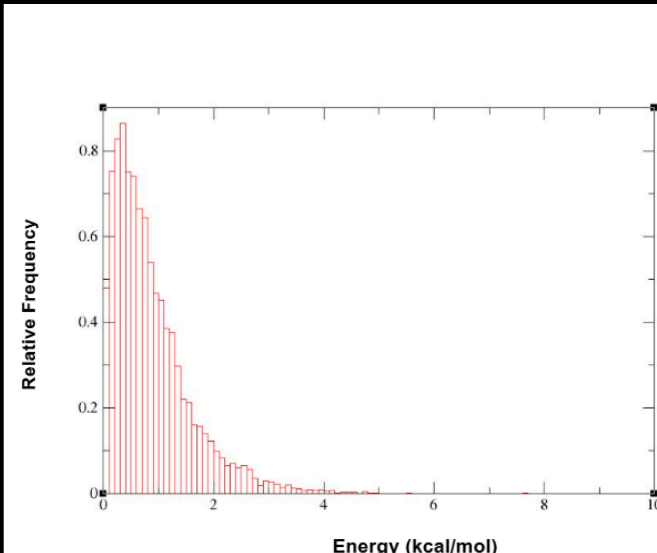
The **N-C terminal separation** rises from **40.7 Å** to **45.1 Å** over **35 time steps**, indicating a net elongation of **4.4 Å**. The trajectory of ubiquitin under constant force pulling shows a stepwise increase in the N-C terminal separation. For constant velocity SMD, applied force peaks at **2636 pN** near timestep **4000**, then fluctuates with unfolding events, and drops to **318 pN** by timestep **14000**.

ROOT MEAN SQUARE DEVIATION (RMSD)



Per-residue RMSD analysis of ubiquitin identified which segments were structurally stable and which were flexible (e.g., N- and C-termini showing **>1.5 Å RMSD**). This methodology can be applied to track localized fluctuations at the surface.

MAXWELL-BOLTZMAN KE DISTRIBUTION

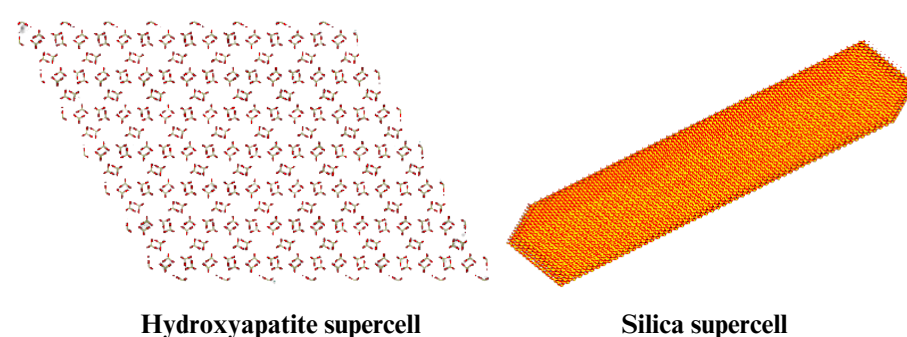


90% of ubiquitin atoms have kinetic energies **<2 kcal/mol** with the frequency sharply decreasing at higher energies.

Maxwell-Boltzmann distribution describes particle energy spread at thermal equilibrium.

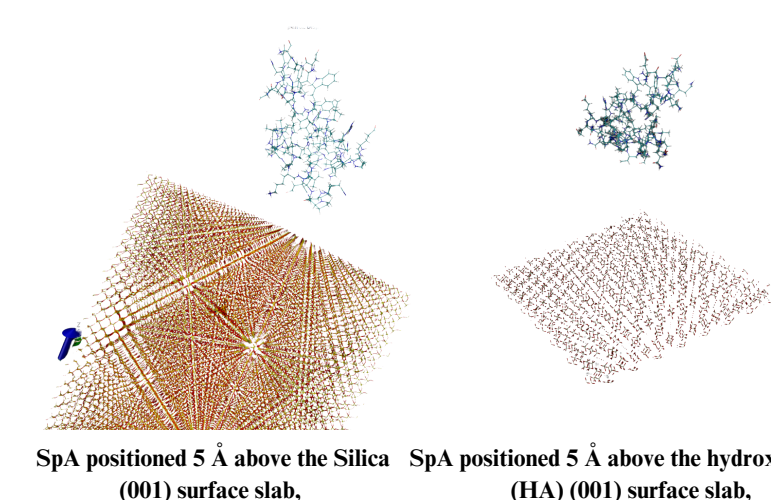
SpA ON HYDROXYAPATITE AND SILICA

Hydroxyapatite (HA) and crystalline silica (α -quartz) surfaces were generated using **Pymatgen** and **ASE**. For HA, a unit cell was cleaved along the **(001)** plane and expanded to a **8 x 7 x 2 supercell** stability and realistic interaction. For α -quartz silica, a trigonal SiO₂ unit cell was expanded into a **14x14x56 supercell**, for large coverage.



Hydroxyapatite supercell

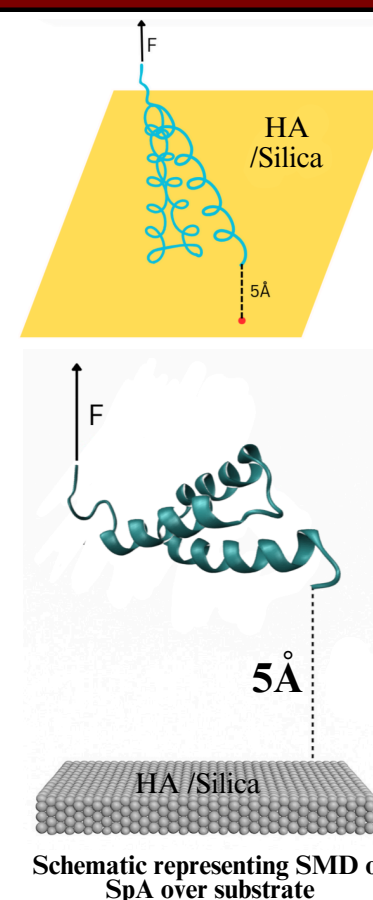
Silica supercell



SpA positioned 5 Å above the Silica (001) surface slab, SpA positioned 5 Å above the hydroxyapatite (HA) (001) surface slab,

The merged **SpA-HA**, **SpA-Silica** system (PDB + CIF) were successfully constructed. However, PSF generation via **psfgen** was hindered by the absence of CHARMM-compatible **interface files** for the substrates. As these topology and parameter files for inorganic-biomolecular interfaces were not publicly available, their unavailability within the limited 2-month timeframe posed a critical bottleneck to proceeding with full-scale simulations.

CONCLUSION AND FUTURE GOALS



Via ubiquitin simulations, **reliability** of the approach for future protein-surface simulations was confirmed and also paved the way for the ultimate objective. The goal is to investigate protein adhesion on biomaterial surfaces through steered molecular dynamics (SMD). The SpA is first positioned **5Å** above HA to **prevent steric clashes** while **enabling nonbonded interactions** to develop during minimization. One side of the protein is subsequently lifted while the substrate atoms remain secured, **offering a stable reference**. This arrangement will allow accurate assessment of **adhesion strength** and **unfolding characteristics**, mimicking the detachment of proteins from material surfaces when subjected to stress.

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